

Results: Experiment 1, H4 ranking clause, RCAEval RE2

Author: Abhineet Asthana. Executed by Claude Code, 2026-08-20. Pre-registration frozen at b0db614791c597dd7b0e91358bf7079762d4ddeb. Analysis code frozen at 7cc789688b8ae8296b144d9daaded23e4d35a67e. Canonical set at 61cec7f57a256cf99ca89d349ae59cb4a685d7c7, revision "c".

Evidence grading: Strong means direct measurement against primary data. Indicative means the evidence points this way but is not conclusive. Asserted means judgement, not established.

ABSTRACT

The result is NULL. The method did not beat the naive control. It performed significantly worse than it. [Strong.]

McNemar exact, paired top-1, n = 180

b (graph method correct, naive wrong) = 12
c (graph method wrong, naive correct) = 53

one-sided p (method greater) = 1.000000
two-sided p = 0.00000028
one-sided p (method LESS) = 0.00000014

VERDICT: FAIL TO REJECT the null. The method underperformed the naive rank-by-worst-symptom control, and the underperformance is itself statistically significant.

What was tested. This tests a **RANKING EXTENSION** introduced in the experiment brief. It is **NOT canonical H4**. Canonical H4 states "Propagated loss differs from node-local sum" with kill condition "Propagated and naive totals agree". That concerns monetary totals. RCAEval contains no revenue, loss or switching cost data, so **the canonical H4 kill condition cannot be evaluated here at all**. Neither this null nor any positive result may be reported as bearing on canonical H4. See pre-registration section 1.3. [Strong.]

Prior art. H4 sits on the RISK AXIS, which Canonical Thesis v2.1c section 5.4 and Part IX Mathematics v3.1c section 9.3.9 grade as **OCCUPIED** by a granted patent family with a **12 March 2018 priority date** (US 10,257,219 and four continuations). That family builds a discovered dependency graph, propagates a seeded disruption by Monte Carlo with a per-edge conditional probability, and reports a dollar-denominated loss exceedance curve. **A positive result here would have demonstrated computability and comparative ranking performance and would have established no novelty. The null establishes even less.** [Strong.]

Power. The study was declared UNDERPOWERED in advance, in pre-registration section 12.5, before any scoring code existed. The design required the method to solve 26 of 34 winnable hard cases, 76.5 percent, against a published state of the art of 21 percent on hard benchmarks. A null was preregistered as the likely outcome. It is the outcome. [Strong.]

1. Headline numbers

TOP-1, node level, n = 180	COUNT	RATE	WILSON 95% CI
NAIVE CONTROL	142/180	78.9%	[72.4%, 84.2%]
Method, PRIMARY S_graph	101/180	56.1%	[48.8%, 63.2%]
TOP-3, node level, n = 180	COUNT	RATE	WILSON 95% CI
NAIVE CONTROL	165/180	91.7%	[86.7%, 94.9%]
Method, PRIMARY S_graph	124/180	68.9%	[61.8%, 75.2%]

The confidence intervals do not overlap at top-1 or top-3. [Strong.]

Of the 162 cases where the root cause was reachable at all, the method ranked it WORSE than the naive control in 53 cases, BETTER in 17, and EQUAL in 92. [Strong.]

Run integrity: 180 of 180 cases completed, **0 errors**, maximum Monte Carlo standard error **0.0011**, total runtime 699 seconds. [Strong.]

2. Scope and exclusions

2.1 n = 180, not 270

DATASET	CASES	STATUS
RE2-OB	90	INCLUDED
RE2-TT	90	INCLUDED
RE2-SS	0	EXCLUDED. Sock Shop contains ZERO trace files. Verified by count: 90 case directories, 0 traces.csv, 0 tracets_*.csv. A trace-mined method cannot run on it. Stated limitation, not a silent omission.
TOTAL	180	

One further exclusion: RE2-OB/checkoutservice_cpu/multi-source-data, a byte-identical duplicate of checkoutservice_cpu/1 (md5 956052e9c07123b4517e37a0b536d9cf). Not a case. [Strong.]

2.2 n by system and fault type

SYSTEM	cpu	delay	disk	loss	mem	socket	TOTAL
RE2-OB	15	15	15	15	15	15	90
RE2-TT	15	15	15	15	15	15	90
TOTAL	30	30	30	30	30	30	180

3. The 18 zeros, as a line item

```
+-----+
| RE2-TT cases where the annotated root cause is ABSENT from |
| the mined graph, and which therefore score ZERO under the |
| PRIMARY candidate set:                                18 of 180 |
|                                                         |
| All 18 are ts-auth-service.                             |
+-----+
```

ts-auth-service emits roughly 10,000 spans per case. Every one is either a trace root (null parent) or parented by another ts-auth-service span, so it has no cross-service edge and Appendix A.1's mining never adds it as a node. It is traced in 90 of 90 TT cases and absent from the mined graph in 90 of 90. [Strong.]

These 18 zeros are the honest coverage cost of a mined-substrate method, not a scoring artefact. What the graph cannot see, the graph cannot blame. They are reported as a first-class part of the result, and the naive control scores 14 of those 18 correctly. [Strong.]

4. Primary result, in full

4.1 Component level versus node level, reported separately

```
+-----+-----+-----+
| PRIMARY S_graph, n = 180 | COMPONENT | NODE |
+-----+-----+-----+
| top-1                    | 101/180 | 101/180 |
| top-3                    | 124/180 | 124/180 |
+-----+-----+-----+
| Cases where the root cause sat inside a multi-node SCC |
| and so could not be resolved internally:          15 |
+-----+-----+-----+
```

No accuracy came from coarser granularity. Component-level and node-level top-1 are identical at 101. In the 15 cases where the root cause sat inside a multi-node component, the worst-rank tie rule pushed the node rank below the component rank, and none of those 15 was a top-1 hit at either level. [Strong.]

4.2 Primary statistic versus S_full

```
+-----+-----+-----+-----+
| STATISTIC                | TOP-1 | TOP-3 | STATUS |
+-----+-----+-----+-----+
| S_graph (PRIMARY)        | 101/180 | 124/180 | Seed coordinate |
|                          | 56.1%  | 68.9%  | EXCLUDED. Non-circular. |
|                          |         |         | This is the tested |
|                          |         |         | statistic.          |
+-----+-----+-----+-----+
| S_full (SECONDARY)       | 117/180 | 143/180 | CONTAINS THE NAIVE |
|                          | 65.0%  | 79.4%  | CONTROL. NOT tested |
|                          |         |         | against it. Reported |
|                          |         |         | for completeness only. |
+-----+-----+-----+-----+
```

S_full scores 16 cases higher than S_graph. That gap is the naive control leaking back into the statistic, which is precisely what the circularity guard was preregistered to remove. It is not evidence of graph performance. [Strong.]

4.3 Candidate sets

CANDIDATE SET	S_graph TOP-1	S_graph TOP-3	S_full TOP-1	S_full TOP-3
PRIMARY, graph nodes only	101/180 56.1%	124/180 68.9%	117/180 65.0%	143/180 79.4%
SECONDARY, all traced svcs UPPER BOUND ONLY	98/180 54.4%	123/180 68.3%	124/180 68.9%	156/180 86.7%

CAPTION, required on the secondary row. The all-traced-services figure is an UPPER BOUND. Admitting graph-isolated candidates lets the method fall back to naive-control behaviour exactly where the graph failed, importing naive performance into the graph method's score. It is never reported alone.

A correction to that caption's own reasoning, logged as D-03. The described mechanism is real for S_full and NOT real for S_graph. Under S_graph the seed's coordinate is excluded, so a graph-isolated candidate has an all-zero predicted vector over the remaining services, its correlation is undefined, and it ranks last. Measured directly: under the secondary set, the 18 ts-auth cases score **0 of 18** on S_graph and **7 of 18** on S_full. The secondary set therefore lowers S_graph slightly, from 56.1 to 54.4 percent, by adding competitors, and raises S_full, from 65.0 to 68.9 percent, by the naive fallback. [Strong.]

Neither candidate set beats naive:

McNEMAR vs NAIVE	b	c	one-sided p
PRIMARY S_graph	12	53	1.000000
SECONDARY S_graph	12	56	1.000000
SECONDARY S_full (not valid as a test; shown for record)	13	31	0.998171

4.4 Decomposition of the 53 losses

c = 53 cases the method lost that naive won	COUNT
Root cause absent from mined graph (ts-auth)	14
Root cause in the graph, method still missed	39

Even discounting every unreachable case, the method lost 39 cases the naive control won, against 12 it won. The ts-auth coverage gap is not the explanation for the null. [Strong.]

5. Hard versus easy

Classified before scoring, per Appendix A.4, on `latency-90`. Under this definition the naive control is correct on exactly the easy cases and wrong on exactly the hard ones, so naive top-1 and the easy rate are the same number by construction.

SYSTEM	n	EASY	HARD
RE2-OB	90	72 (80.0%)	18 (20.0%)
RE2-TT	90	70 (77.8%)	20 (22.2%)
COMBINED	180	142 (78.9%)	38 (21.1%)

SUBSET	n	METHOD TOP-1 (node, S_graph)	METHOD TOP-3 (node, S_graph)	WILSON 95% CI on TOP-1
HARD	38	12 (31.6%)	17 (44.7%)	[19.1%, 47.5%]
EASY	142	89 (62.7%)	107 (75.4%)	[54.5%, 70.2%]

Two things must be said about this table together, because either alone misleads.

First, the method does something on hard cases that the naive control by construction cannot. 31.6 percent top-1 on hard cases, where naive scores 0 percent by definition. The point estimate exceeds the 21 percent that Fang et al. report as state of the art on a hard benchmark, though the confidence interval [19.1%, 47.5%] contains 21 percent, so this is **[Indicative]**, not Strong.

Second, it destroys easy cases to get there. On the 142 easy cases the method scores 62.7 percent where naive scores 100 percent by definition. It converts 53 naive wins into losses to gain 12 hard wins. The trade is strongly negative. [Strong.]

A method that only works on easy cases has shown nothing. A method that wins some hard cases by losing four times as many easy ones has also shown nothing about the ranking clause. [Asserted, this is a judgement.]

6. Baselines

METHOD	TOP-1	RATE	WILSON 95% CI	RAN ON
NAIVE CONTROL	142/180	78.9%	[72.4%, 84.2%]	180/180
nsigma	130/180	72.2%	[65.3%, 78.2%]	180/180
BARO	127/180	70.6%	[63.5%, 76.7%]	180/180
METHOD S_graph	101/180	56.1%	[48.8%, 63.2%]	180/180
epsilon-Diagnosis	11/180	6.1%	[3.4%, 10.6%]	27/180
Random floor	11/180	6.1%	[3.4%, 10.6%]	180/180
CIRCA	6/180	3.3%	[1.5%, 7.1%]	180/180
RCD	0/180	0.0%	[0.0%, 2.1%]	0/180

METHOD	TOP-3	RATE	WILSON 95% CI
NAIVE CONTROL	165/180	91.7%	[86.7%, 94.9%]
BARO	163/180	90.6%	[85.4%, 94.0%]
nsigma	162/180	90.0%	[84.7%, 93.6%]
METHOD S_graph	124/180	68.9%	[61.8%, 75.2%]
Random floor	30/180	16.7%	[11.9%, 22.8%]
CIRCA	30/180	16.7%	[11.9%, 22.8%]
epsilon-Diagnosis	27/180	15.0%	[10.5%, 20.9%]
RCD	0/180	0.0%	[0.0%, 2.1%]

Paired McNemar, method against each baseline, one-sided, method greater:

COMPARISON	b	c	p one-sided	HOLM-ADJUSTED
vs NAIVE (PRIMARY)	12	53	1.000000	not corrected
vs nsigma	24	53	0.999731	1.000000
vs BARO	25	51	0.999119	1.000000
vs epsilon-Diag	92	2	< 1e-6	< 1e-6
vs CIRCA	98	3	< 1e-6	< 1e-6
vs RCD	101	0	< 1e-6	< 1e-6
vs Random floor	95	5	< 1e-6	not in family

Three of those five wins are not findings and must not be read as such. [Strong.]

- **RCD produced no output on any case.** RCAEval 1.6.0's `e2e/rcd.py` calls `SkeletonDiscovery.local_skeleton_discovery` on its default path and passes a `labels=` keyword to `skeleton_discovery` on the other. Neither exists in causal-learn; checked at 0.1.3.3, 0.1.3.6, 0.1.3.8 and 0.1.4.8. Both are present only in the RCD authors' fork, which RCAEval does not pin. 130 cases failed with the `AttributeError`, 50 with a downstream `NaN` error. **Beating a baseline that did not run is not a result.** RCAEval's internals were not patched to manufacture a number. Logged as D-04.
- **epsilon-Diagnosis ranks at most three services** (median 3, maximum 3, and 0 in seven cases). Its output is structurally incapable of ranking beyond top-3, so its low score reflects coverage, not only accuracy.
- **CIRCA scored below the random floor** (3.3 percent against 6.1 percent) under RCAEval's default settings. Reported as measured. No attempt was made to tune it, since tuning a baseline after seeing results is exactly what pre-registration forbids. [Strong.]

The finding that matters in this table is not about the method at all. The naive rank-by-worst-symptom control beat **every** RCAEval baseline, including BARO and nsigma, the two that ran cleanly on all 180 cases. Part IX Mathematics v3.1c section 9.11 warned, before this experiment was designed, that "several widely used benchmarks are simple enough that trivial rule-based methods match state-of-the-art results, so choose a hard one". On RE2 a trivial rule does not merely match the baselines. It beats all of them. [Strong.]

7. Sensitivity grid

Nine cells, full 180 cases each, 1,620 runs, 0 errors.

CELL	TOP-1	RATE	TOP-3	b	c	p ONE-SIDED
a1 b1 p_base 0.35	103	57.2%	124	12	51	1.0000
a2 b5 p_base 0.35	101	56.1%	124	12	53	1.0000
a1 b1 p_base 0.75	100	55.6%	123	12	54	1.0000
a5 b2 p_base 0.75	100	55.6%	123	12	54	1.0000
a2 b5 p_base 0.75	98	54.4%	124	12	56	1.0000
a5 b2 p_base 0.35	97	53.9%	123	12	57	1.0000
a1 b1 p_base 0.55	96	53.3%	123	12	58	1.0000
a2 b5 p_base 0.55	96	53.3%	124	12	58	1.0000
a5 b2 p_base 0.55	96	53.3%	124	12	58	1.0000
NAIVE CONTROL (fixed)	142	78.9%	165			

top-1 range: 53.3% to 57.2%. Spread 3.9 percentage points.
Cells reaching $p \leq 0.05$: 0 of 9
Cells beating naive at all ($b > c$): 0 of 9
b was exactly 12 in all nine cells.

Appendix A.2 required that if top-1 accuracy moves more than a few points across the grid, the parameter choice rather than the graph is driving the result, and that this must be the headline. **It does not.** The spread is 3.9 points, and the gap to the naive control is 21.7 to 25.6 points in every cell. The null is not a parameter artefact. [Strong.]

The four ASSERTED defaults (a=2, b=5, p_base=0.55, p_shared=0.40) remain unfitted and unjustified. The grid shows the conclusion does not depend on them, which is a different and weaker claim than the defaults being correct. [Strong.]

8. Monte Carlo seed sensitivity, an unplanned finding

The default parameter cell was run twice under different derived seeds, because the two runners built the cell identifier differently ("a2.0_b5.0_..." against "a2_b5_..."). This produced an accidental replication.

```
+-----+-----+
| Same parameters, two seeds | TOP-1 |
+-----+-----+
| seed set A (full_default) | 101 |
| seed set B (grid)         | 96 |
+-----+-----+
| Node ranks differed on 15 of 180 cases (8.3%). |
| Every difference was exactly +/- 1 rank.       |
| Top-1 flips: 1 gained, 6 lost.                 |
+-----+-----+
```

At $n_{sim} = 200,000$ the maximum Monte Carlo standard error is 0.0011, which is small in absolute terms but sufficient to reorder near-tied candidates. **Top-1 carries roughly plus or minus 3 percentage points of seed jitter.** [Strong.]

This does not affect the conclusion: the gap to naive is 22.8 points, roughly seven times the jitter. It is reported because a point estimate without its variability is exactly what the framework's distribution rule forbids, and because it was not anticipated in the pre-registration. [Strong.]

9. Every failure, with a cause

79 of 180 cases were top-1 failures for the method.

```
+-----+-----+
| FAILURE CATEGORY | COUNT |
+-----+-----+
| D. Ranked, but not first | 46 |
| A. Root cause absent from the mined graph | 18 |
|   (all ts-auth-service, scored zero) |
| B. Root cause inside a multi-node SCC, so | 15 |
|   unresolvable at node level |
+-----+-----+
```

Category A, 18 cases. Cause established, not guessed: ts-auth-service has no cross-service parent-child edge in any case, so A.1 mining cannot place it in the graph. [Strong.]

Category B, 15 cases. Cause established: Tarjan condensation merges services in a cycle into one component, and the preregistered worst-rank tie rule then assigns every member the worst rank in the block. The cycles observed are small, chiefly (ts-travel2-service, ts-seat-service) and (ts-seat-service, ts-travel-service). This is the cost of the cycle handling and it was accepted in advance. [Strong.]

Category D, 46 cases. Rank distribution:

RANK		CASES
2	17	
3	6	
4	12	
5	9	
10	1	
19	1	

The naive control got 35 of these 46 right. **[Indicative]** cause: the graph-consistency score ranks by how well the observed degradation pattern of OTHER services matches propagation from the candidate. In Online Boutique the mined graph has only 7 nodes and 9 edges, so many candidates induce near-identical ancestor sets and the score has little to separate them. 17 of the 46 are rank 2, consistent with the true cause being a close second on a statistic with coarse resolution rather than being badly misplaced.

Per fault type and system:

METHOD	cpu (n=15)	delay (n=15)	disk (n=15)	loss (n=15)	mem (n=15)	socket (n=15)
RE2-OB	10	11	10	12	10	7
RE2-TT	6	9	5	7	7	7
NAIVE	cpu	delay	disk	loss	mem	socket
RE2-OB	11	11	13	13	13	11
RE2-TT	13	13	9	8	12	15

By system:

SYSTEM	n	METHOD TOP-1	NAIVE TOP-1
RE2-OB	90	60 (66.7%)	72 (80.0%)
RE2-TT	90	41 (45.6%)	70 (77.8%)

The method is markedly worse on Train Ticket, which is the larger, cyclic, lower-coverage system. It is also, per pre-registration section 2.2, the only system in RE2 capable of testing the mined-substrate claim at all, because the Online Boutique graph is a constant. **The method performs worst exactly where the claim would have to be tested.** [Strong.]

10. Monotonicity: UNVERIFIED

Part IX Mathematics v3.1c requires a monotonicity check before fault-tree logic: "Retries, fallbacks, circuit breakers, load shedding and graceful degradation all break monotonicity."

Retry-shaped probe, fraction of parented spans sharing a (traceID, parentSpanID, operationName) key:

SYSTEM	MIN	MEDIAN	MAX
RE2-OB	28.5%	30.0%	31.2%
RE2-TT	23.1%	29.8%	40.6%

Roughly 30 percent of parented spans carry a retry-shaped signature. [Strong.]

VERDICT: the monotonicity precondition is UNVERIFIED and is not upgraded. Circuit breakers, fallbacks and graceful degradation leave no record in these traces: a span never emitted because a breaker was open cannot be counted. This experiment therefore cannot establish that the monotone assumption holds, and the propagation model embeds it. Per the framework's rule that a recorded refusal beats a number nobody should trust, this is recorded as a refusal. [Strong.]

The probe finding and the verdict are reported together deliberately: a 30 percent retry rate is a reason to doubt monotonicity, not a measurement of it.

11. What this result licences, and what it does not

11.1 Licensed

- On RE2-OB and RE2-TT, graph propagation with common-cause conditioning, scored by a non-circular graph-consistency statistic, **localises annotated root causes less accurately than ranking services by worst observed latency degradation**. 56.1 percent against 78.9 percent top-1, McNemar $b=12$ $c=53$, two-sided $p = 0.00000028$. [Strong.]
- The result is stable across the preregistered nine-cell parameter grid, spread 3.9 points, and is not a parameter artefact. [Strong.]
- On this benchmark a trivial rule-based control beats every RCAEval baseline tested. [Strong.]

11.2 NOT licensed

- **Nothing about canonical H4.** Canonical H4 concerns monetary totals and its kill condition cannot be evaluated on RCAEval. This null does not kill canonical H4 and a positive would not have supported it. [Strong.]
- **Nothing about novelty**, in either direction. The risk axis is OCCUPIED by a granted family with March 2018 priority. [Strong.]
- **Nothing about the economic thesis.** No monetary quantity was computed at any point. [Strong.]
- **Nothing conclusive about propagation modelling in general.** This tests one specification of one model on one benchmark that the framework's own Part IX warned was likely too easy. A better propagation method may exist. This experiment did not find one and did not look for one. [Asserted.]

11.3 The honest reading

[Asserted, and it is a judgement rather than a measurement.] The most defensible thing this experiment produced is not the null on the ranking extension. It is the measurement that a one-line rule scores 78.9 percent on a benchmark used to evaluate published RCA methods, and beats all of them. That is a finding about RE2, and it reproduces Fang et al. and vindicates the warning already written into Part IX section 9.11 before this experiment began.

The pre-registration predicted this outcome in section 12.5, before any code existed, on arithmetic alone. The study was underpowered by a factor of roughly three and a half against published state of the art on hard cases. It returned the predicted null. The design was not adjusted at any point to change that.

12. Deviations encountered during the run

Recorded in full in PREREGISTRATION.md section 16. Summary:

+-----+		
D-01	Correction IDs renumbered (R13-R16 -> R17-R20, T-M/T-N -> T-T/T-U) after the first assignment was made against a stale canonical set. Documentation only.	
+-----+		
D-02	Canonical citations refreshed to revision "c". H4 wording verified unchanged; E3 verified OCCUPIED.	
+-----+		
D-03	The frozen runner did not implement the preregistered SECONDARY candidate set. Added in a separate file rather than by editing frozen code. Includes a correction to section 9.2's own reasoning about S_graph.	
+-----+		
D-04	RCD is not runnable against any released causal-learn. Reported as such rather than patched around.	
+-----+		

No deviation changed the hypothesis, the metric, the statistical test, the exclusions, the candidate sets, the tie-breaking, or the power analysis. [Strong.]

13. Correction log

Nine corrections and one standing method rule are recorded in `docs/CORRECTION_LOG.md`, with formal IDs assigned in the framework's convention against the v2.1c canonical set:

+-----+		
R17	Prior run: root cause presented as discovered when it was a label lookup	
R18	Prior run: headline number taken from a source nobody opened. The x4.08 figure came from tracets_lat.csv, whose semantics are undocumented and unreproducible. Recomputed on canonical latency-90, that case shows NOTHING degraded (all services x0.95 to x1.06).	
R19	Prior run: comparative claim true only under an undisclosed restriction to the 7 traced services	
R20	Prior run: showcase case unrepresentative, one of the five weakest of ninety and one of 18 hard cases	
T-T	Transmission probability unbounded above in A.2	
T-U	Ranking statistic contains the control it is compared against	
M-01	STANDING RULE. A file retrieved through a representation layer is not the file. Byte-exact or it does not count.	
+-----+		

R18, the patent citation error corrected at v1.1c, and M-01 share one shape: a source taken at one level of detail and carried forward as established.

14. Reproduction

Pre-registration	b0db614791c597dd7b0e91358bf7079762d4ddeb
Analysis code	7cc789688b8ae8296b144d9daaded23e4d35a67e
Canonical set	61cec7f57a256cf99ca89d349ae59cb4a685d7c7
Secondary runner	3ea7a32e9d4486478a4661e062941a952c4f50b3
Master seed	20260820
Per-case seeds	SHA256(case_id cell_id) truncated to 32 bits, recorded per row in the results files
Data	Zenodo record 14590730, RE2-0B.zip and RE2-TT.zip
Environment	Python 3.12.3, RCAEval 1.6.0, numpy 1.26.4, pandas 3.0.5, networkx 3.6.1, causal-learn 0.1.4.8
Outputs	results/full_default.csv 180 rows, primary results/grid.csv 1,620 rows, nine cells results/secondary.csv 180 rows, upper bound results/baselines.csv 180 rows, six baselines results/grid_summary.csv 9 rows

The primary analysis uses deterministic worst-rank tie-breaking, so no seed affects the ranking rule. Seeds affect only the Monte Carlo propagation estimates, whose sensitivity is quantified in section 8.